

FARMINGDALE REPUBLIC, NY, USA

WMO: 744864

Lat: 40.734N

Lon: 73.417W

Elev: 81

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54787

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.2	(c) 16.5	(d) -6.2	(e) 4.0	(f) 15.4	(g) -1.4	(h) 5.1	(i) 20.2	(j) 27.2	(k) 37.0	(l) 25.0	(m) 34.9	(n) 10.3	(o) 310	(p) 0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	89.8	73.8	86.4	72.1	83.7	71.2	76.8	84.3	75.5	81.8	74.3	80.2	11.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	130.4	80.3	73.3	124.2	78.6	72.4	120.5	77.9	40.2	84.3	39.0	82.2	37.8	79.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.4	(b) 20.8	(c) 18.7	(e) 7.3	(f) 94.6	(g) 4.9	(h) 3.7	(i) 3.8	(j) 97.3	(k) 1.0	(l) 99.5	(m) -1.8	(n) 101.6	(o) -5.3	(p) 104.3		
(4) 24.4	(5) 20.8	(6) 18.7	(7) 7.3	(8) 94.6	(9) 4.9	(10) 3.7	(11) 3.8	(12) 97.3	(13) 1.0	(14) 99.5	(15) -1.8	(16) 101.6	(17) -5.3	(18) 104.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	32.1	34.4	40.5	50.5	60.0	69.3	75.4	74.4	67.7	56.7	46.2	37.9
	DBStd	16.66	9.57	8.18	8.07	6.84	6.54	6.29	4.87	4.78	5.86	7.46	7.63	8.22
	HDD50	1948	556	440	309	73	2	0	0	0	0	23	161	384
	HDD65	5008	1019	858	759	439	185	30	0	1	39	273	565	840
	CDD50	3361	2	2	16	86	313	579	789	756	531	232	46	9
	CDD65	946	0	0	0	3	31	159	324	292	120	17	0	0
	CDH74	6217	0	0	3	34	198	982	2466	1960	521	53	0	0
	CDH80	1666	0	0	0	8	38	264	761	507	80	8	0	0
(14) Wind	WSAvg	9.0	9.7	9.8	10.4	9.9	8.8	8.4	8.3	7.7	8.1	8.7	9.0	9.3
(15) Precipitation	PrecAvg	44.2	3.1	3.2	4.3	3.7	3.5	4.0	3.5	4.2	3.2	3.6	3.5	4.1
	PrecMax	50.4	4.9	5.7	8.3	6.5	4.9	9.6	5.9	11.3	8.0	15.5	5.2	6.5
	PrecMin	35.1	1.9	1.2	1.6	1.5	1.2	1.1	1.5	1.7	1.1	0.0	1.3	2.0
	PrecStd	5.4	0.9	1.3	1.9	1.5	1.1	2.4	1.2	2.3	1.9	3.4	1.1	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	60.3	68.6	79.1	84.3	90.7	95.0	91.8	86.5	80.1	67.5	61.6
		MCWB	51.9	51.3	55.9	60.8	68.1	73.6	75.3	75.5	72.3	67.2	58.6	55.9
	2%	DB	53.6	53.9	60.9	70.1	79.6	86.4	90.4	88.2	81.9	73.7	63.5	56.8
		MCWB	50.7	47.3	50.3	56.5	65.8	71.4	73.9	73.3	69.6	66.2	57.8	54.1
	5%	DB	49.8	49.8	55.6	65.2	74.9	82.5	87.1	85.0	79.3	71.3	61.2	54.0
		MCWB	46.0	45.4	48.6	53.7	63.8	69.9	73.0	71.5	68.9	64.6	56.9	51.7
	10%	DB	46.0	46.2	52.1	61.3	71.1	79.6	84.0	82.3	76.9	68.1	58.2	50.9
		MCWB	42.1	41.6	46.1	51.4	61.3	68.2	71.8	71.3	68.3	62.2	53.8	47.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	53.8	58.5	64.4	71.6	76.6	78.9	78.7	75.9	71.3	63.0	58.3
		MCDB	55.6	56.3	67.4	74.6	81.6	86.3	89.7	87.4	81.2	74.4	65.0	59.9
	2%	WB	51.2	49.6	53.5	59.6	68.6	74.3	76.9	76.9	74.2	68.6	60.3	54.7
		MCDB	52.6	52.3	57.2	66.1	75.6	82.3	85.2	83.3	78.4	72.5	62.5	56.1
	5%	WB	46.5	45.9	50.1	56.6	66.2	72.5	75.6	75.5	72.6	66.2	57.8	52.2
		MCDB	48.7	48.3	54.4	61.4	72.6	79.5	82.7	81.0	76.5	69.8	60.1	54.0
	10%	WB	42.7	42.2	46.8	53.9	63.7	70.6	74.4	74.3	70.9	63.7	54.9	47.7
		MCDB	45.6	46.1	50.9	58.4	68.6	76.7	80.9	79.4	74.8	66.9	57.8	50.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	14.0	14.4	15.5	15.1	14.7	14.2	13.6	14.1	14.3	13.8	12.3
		MCDBR	16.7	18.2	19.3	21.6	20.6	18.2	17.3	15.9	15.3	16.4	15.4	15.1
		MCWBR	15.0	14.3	12.5	11.7	11.0	8.7	6.8	6.7	8.1	10.1	12.6	15.1
	5% WB	MCDBR	16.1	17.7	17.9	17.7	18.1	15.9	14.1	12.5	11.9	13.6	13.8	14.5
		MCWBR	16.0	16.3	14.0	12.0	11.0	8.8	7.0	6.6	8.1	10.5	13.1	15.7
(40) Clear-Sky Solar Irradiance	taub		0.310	0.319	0.344	0.382	0.430	0.480	0.489	0.471	0.399	0.351	0.323	0.311
	taud		2.503	2.483	2.459	2.379	2.274	2.173	2.158	2.203	2.377	2.524	2.554	2.553
	Ebn at Noon		271	285	288	283	270	256	252	253	266	270	264	262
	Edh at Noon		24	28	32	37	42	47	47	44	34	27	23	21
(44) All-Sky Solar Radiation	RadAvg		589	875	1204	1544	1722	1890	1914	1708	1394	965	671	505
	RadStd		49	87	102	153	171	178	95	109	115	92	57	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+115

Nomenclature: See separate page